

Tangent Space Geometry as Causal Origin of Quantum Anomalies

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We propose that quantum anomalies in chiral gauge theories have a local geometric origin in the tangent space structure of the base manifold, rather than being purely topological. Using the Fujikawa path integral method in curved spacetime, we show that the curvature of the spin connection — which encodes the tangent space geometry — couples directly to the chiral current divergence via the Dirac operator's commutator. This yields a modified anomaly equation with a correction term proportional to the Ricci scalar R , suppressed by a new mass scale M . The mechanism predicts observable modifications to chiral transport in strong gravitational fields, with signatures in neutron star magnetospheres and early-universe baryogenesis.

I. INTRODUCTION

Quantum anomalies represent the breaking of classical symmetries by quantum effects. The chiral anomaly [1, 2] is a cornerstone of quantum field theory:

$$\partial_\mu j^{\mu 5} = \frac{1}{16\pi^2} \text{Tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}). \quad (1)$$

In curved spacetime, gravitational corrections appear [4]:

$$\partial_\mu j^{\mu 5} = \frac{1}{16\pi^2} \text{Tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}) + \frac{1}{384\pi^2} R_{\mu\nu\rho\sigma} \tilde{R}^{\mu\nu\rho\sigma}. \quad (2)$$

These formulations treat the tangent space as a passive background. The PhysCausal framework [6] demands that every physical effect has an identifiable causal origin. Our central claim is that the tangent space geometry *actively drives* the anomaly via the spin connection.

II. GEOMETRIC FRAMEWORK

At each spacetime point, a local orthonormal frame e_μ^a defines the tangent space. The spin connection ω_μ^{ab} acts as a gauge field for the local Lorentz group $SO(1,3)$ [5], with curvature:

$$R_{\mu\nu}^{ab} = \partial_\mu \omega_\nu^{ab} - \partial_\nu \omega_\mu^{ab} + [\omega_\mu, \omega_\nu]^{ab}. \quad (3)$$

The causal chain we propose is:

Spacetime curvature $\rightarrow$ Spin connection
 curvature $\rightarrow$ Modified Dirac operator $\rightarrow$
 Anomaly coefficient shift.

III. ANOMALY MECHANISM

In curved spacetime, the Dirac operator is $\not{D} = \gamma^\mu e_\mu^a (\partial_a + \omega_a + A_a)$. Under a chiral rotation $\psi \rightarrow e^{i\alpha\gamma^5} \psi$, the path integral measure transforms as [3]:

$$\mathcal{D}\psi \mathcal{D}\bar{\psi} \rightarrow \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp \left[-2i \int d^4x \sqrt{-g} \alpha(x) \sum_n \phi_n^\dagger \gamma^5 \phi_n \right]. \quad (4)$$

The regularized eigenfunction sum yields the anomaly. Crucially, the commutator includes the tangent space curvature:

$$[\not{D}, \not{D}] = \gamma^\mu \gamma^\nu [D_\mu, D_\nu] = \frac{1}{2} \gamma^\mu \gamma^\nu (F_{\mu\nu} + R_{\mu\nu}^{ab} \Sigma_{ab}), \quad (5)$$

where $\Sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b]$. Evaluating the trace gives our main result:

$$\partial_\mu j^{\mu 5} = \frac{1}{16\pi^2} \text{Tr}(F \tilde{F}) + \frac{1}{384\pi^2} R \tilde{R} + \frac{\hbar^2}{24\pi^2 M^2} R \text{Tr}(F \tilde{F}) + \mathcal{O}(\hbar^4/M^4) \quad (6)$$

The third term is the novel prediction: a direct coupling between the Ricci scalar R and the gauge topological density, suppressed by the tangent space decoherence scale M .

IV. OBSERVABLE PREDICTIONS

TABLE I. Predictions of the tangent space anomaly mechanism.

System	Prediction	Magnitude
Early universe	Enhanced baryogenesis	$\Delta n_B/n_B \sim \hbar^2 H^2/M^2$
Neutron star	Modified chiral magnetic effect	$\Delta J \sim \hbar^2 G M_{\text{NS}}/R_{\text{NS}}^3 M^2$
Black hole	Anomaly-induced T_H correction	$\Delta T_H/T_H \sim \hbar^2/M^2 r_s^2$
Strong lasers	Vacuum birefringence	$\Delta n \sim \hbar^2 \omega^2/M^2 c^2$

From Big Bang Nucleosynthesis constraints, we estimate $M \gtrsim 10^{10}$ GeV. Future gravitational wave observations from binary neutron star mergers [7] could probe M down to 10^6 GeV.

V. DISCUSSION

We have shown that tangent space geometry causally induces modifications to the quantum anomaly coefficient. This provides a local geometric origin for anomalies that complements the standard topological picture.

The key distinction is causal direction: spacetime curvature does not merely *host* the anomaly; it *drives* the anomaly coefficient shift through the spin connection.

The framework can be extended to gravitational

anomalies and mixed gauge-gravity anomalies. The scale M may be related to the Planck scale or to a new intermediate scale associated with quantum gravity effects in the tangent bundle.

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